

3. ANALYTICAL SOLUTIONS

Our objective is to build exact solutions that fit, in particular, the physiological problem under study,

$$\tau \partial_t u - \alpha \Delta u + Bu = f, \quad (12)$$

via the separable variable method, in cylindrical coordinates *i.e.* at the position (r, z) and the time t . Using Bernoulli–Fourier technique, we have

$$(\tau X'(t) + [-\alpha(\beta + \eta^2) + B]X(t)) R(r)Z(z) = f(r, z, t), \quad (13)$$

where the constants $\beta, \eta \in \mathbb{R}$ are arbitrary,

Radiative transfer: $\tau = 1/\nu$, $\alpha = D$, $B = \mu_a$ and $f = S$ in Subsection 2.1;

Heat transfer: $\tau = \rho c_p$, $\alpha = k$, $B = c_b \omega_b(t)$ and $f = q$ in Subsection 2.2.

The functions X , Z and R are elementary solutions to the system of ordinary differential equations (ODE)

$$\begin{cases} X'(t) = \zeta X(t) \\ Z''(z) = \eta^2 Z(z) \\ (rR'(r))' = \beta rR(r) \end{cases} \quad (14)$$

for some time parameter $\zeta \in \mathbb{R}$. The nonconstant behavior of ω_b (cf. (11)) does not invalidate the application of this technique in determining an analytical solution for the heat transfer.

Firstly, a particular solution is available due to the Duhamel principle (for details, see [9] and the references therein). Secondly, let us seek for elementary solutions of the ODE system (14).

The first-order ODE in (14) admits the elementary solution

$$X(t) = \exp\left[\int_0^t \zeta(s) ds\right], \quad t > 0. \quad (15)$$

The second-order ODE in (14), with constant coefficients, admits the elementary solutions

$$Z(z) = \exp[\pm \eta z], \quad z \in \mathbb{R}. \quad (16)$$

The second-order ODE in (14), with nonconstant coefficients, admits the Bessel functions of first and second kind and order 0, respectively, $J_0(\sqrt{|\beta|}r)$ and $Y_0(\sqrt{|\beta|}r)$ if $\beta < 0$; or the modified Bessel functions of first and second kind and order 0, respectively, $I_0(\sqrt{\beta}r)$ and $K_0(\sqrt{\beta}r)$ if $\beta > 0$ [33]. If $\beta = 0$, the elementary solutions reduce to $R(r) = \log[r]$ and the unity function.

Then, a general solution solving the homogeneous equation (13) ($f = 0$) is available by the above elementary solutions, if

$$\tau \zeta + B = \alpha(\beta + \eta^2). \quad (17)$$

Finally, we analyze the PDE (12) at the period of time $0 < t_j + t_p < t < t_{j+1}$, for any $j = 0, \dots, N-1$. This case will describe the homogeneous problem, which is governed without source ($f = 0$), at one pulse-to-pulse interval Δt . For the sake of simplicity, we denote the initial instant of time $t_j + t_p$ by t_0 throughout this subsection.